

Noppapadonthon Palawong

Analog / Mixed-Signal Design Engineer

Corvallis, OR | palawonn@oregonstate.edu | (541) 207-4648 | Languages: Thai, English

PROFESSIONAL SUMMARY

M.Eng. Electrical and Computer Engineering candidate focused on analog/mixed-signal IC design, PLL clock generation, CMOS devices, sensor interfaces, and silicon-oriented validation. Hands-on with Cadence Virtuoso/Spectre transistor-level simulation, MATLAB loop/noise analysis, Silvaco ATHENA/ATLAS TCAD, PCB bring-up, and embedded data acquisition. Strong foundation in PLLs, VCOs, op-amps, phase noise/jitter, ADC interfaces, and bench validation.

TECHNICAL SKILLS

Analog/Mixed-Signal: PLLs, VCOs, PFD/charge pumps, passive loop filters, CMOS op-amps, Miller compensation, feedback/stability, phase noise/jitter, thermal/flicker noise, PSD, ADC/sensor interfaces, signal conditioning

EDA & Analysis: Cadence Virtuoso, ADE/Spectre, PSS/Pnoise, DC/AC/transient simulation, LTspice, MATLAB/Simulink, Silvaco ATHENA/ATLAS, TCAD, Vivado

Hardware / Firmware: PCB schematic/layout, Altium, KiCad, oscilloscope, logic analyzer, DDC112 ADC, STM32/ESP32, ESP8266, C/C++, Python, Verilog

EDUCATION

Oregon State University | M.Eng. Electrical and Computer Engineering | GPA: 3.65

Expected May 2026

Relevant Coursework: Analog CMOS IC, Phase-Locked Loops, Semiconductor Process Integration, VLSI Design, Communication Systems

King Mongkut's University of Technology Thonburi | Bachelor's Degree, Electrical Engineering

ENGINEERING EXPERIENCE

Sensing Platform Developer | Oregon State University, Corvallis, OR

Jul 2025 - Present

- Designed and integrated low-noise optical sensing platform using DDC112 precision ADC, STM32/ESP32 acquisition, and analog signal conditioning for real-time measurement.
- Developed C/C++ firmware for ADC control, sensor communication, calibration routines, and data capture; improved repeatability through noise-aware processing and calibration.
- Performed PCB bring-up and bench validation with oscilloscope/logic analyzer; debugged sensor interface, ADC timing, power integrity, and system-level measurement noise.

Hardware Design Intern | NovaGreen Power System, Thailand

Jun 2023 - Sep 2023

- Designed LM2576 buck-boost converter schematic/layout in Altium for embedded power applications; supported component selection, PCB review, bring-up, and board validation.
- Integrated ESP8266 MCU with analog sensors and LCD/display interface; debugged power, sensor, and MCU interfaces to verify stable embedded hardware operation.

SELECTED ANALOG & MIXED-SIGNAL PROJECTS

Low-Power Wideband PLL Clock Generator (6-12 GHz) | Cadence Virtuoso/Spectre, FreePDK45, MATLAB

- Designed transistor-level 45 nm CMOS PLL with PFD, charge pump, passive loop filter, current-starved ring VCO, and divide-by-32 feedback; optimized bandwidth-jitter-power trade-offs for ≤ 1.1 V operation.
- Characterized VCO tuning from 6 GHz at $V_{ctrl}=400$ mV to 12 GHz at $V_{ctrl}=480$ mV; extracted $KVCO=65\text{-}76$ GHz/V and evaluated control-node noise sensitivity from high ring-VCO tuning gain.
- Verified loop stability using $R=65$ ohm, $C1=1$ nF, $C2=2$ pF: 41.9-49.5 MHz bandwidth and ~ 84 deg phase margin; integrated phase noise to 1.314 ps jitter at 6 GHz and 0.770 ps at 12 GHz with 4.64/7.16 mW power.

Two-Stage CMOS Operational Amplifier | Cadence Virtuoso, 0.18 μ m

- Designed and simulated CMOS op-amp with 80.6 dB gain, 47.47 MHz UGF, and 49 deg phase margin; implemented Miller compensation ($C_c=1.5$ pF, $R_z=4.5$ kOhm), verified DC/AC/transient behavior, and achieved ICMR from -0.882 V to +0.757 V.

CMOS Device & Process Simulation | Silvaco ATHENA/ATLAS TCAD

- Simulated twin-well CMOS inverter process including LOCOS isolation, p/n wells, source/drain implants, gate oxide, contacts, and VTC; extracted $V_{th,NMOS}=+1.55$ V and $V_{th,PMOS}=-0.53$ V to connect process variables with switching point, noise margin, leakage, and static power.

Capacitive Sensor Interface & Calibration | ESP32, ADC, Signal Conditioning

- Built 3.3 V / 12-bit ESP32 capacitive sensor signal chain; mapped 3000-to-1200 count dry/wet range with -18 counts/% moisture sensitivity and evaluated temperature, soil type, salinity, placement, and calibration limits.

Analog Communication System (DSB-SC AM) | MATLAB/Simulink

- Simulated modulation/demodulation chain with additive noise; validated ± 25 kHz sidebands, coherent recovery, 4 kHz LPF selection, PSD/SNR behavior, and phase/frequency error effects on recovered signal quality.

FPGA RTL & Mixed-Signal Digital Support | Verilog, Vivado

- Implemented synthesizable RTL support blocks including synchronizers/debouncing, edge detectors, shift registers, DSP/BRAM logic, AES-128 T-table lookup, and asynchronous FIFO with Gray-code CDC.
- Created testbenches, test vectors, waveform checks, and FPGA integration demos for multi-clock digital systems that support mixed-signal control and data movement.